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From experiments to prediction: a machine-learning framework for discovering synergy in complex mixtures

Machine-learning framework

Publication: *Scientific Reports* (2026)

Programming: Python • XGBoost • Statistical modelling

Applications: Drug discovery • Food science • Biotechnology • Agriculture

One analytical framework. Unlimited mixture applications.

I'm pleased to share our latest publication in *Scientific Reports*.

Although this study focused on phenolic antioxidants, the broader goal was to develop a reproducible machine-learning framework for identifying **synergistic, additive and antagonistic interactions in complex compound mixtures**. Phenolic compounds served as the model system, but the workflow is applicable wherever the combined behaviour of multiple compounds can be experimentally measured and modelled.

The challenge wasn't simply measuring antioxidant activity; it was determining whether mixtures behaved as expected or exhibited genuine **synergistic** or **antagonistic** effects.

To address this, I designed the computational workflow and developed the complete Python analysis pipeline, integrating:

- ✓ Exploratory data analysis
- ✓ Regression modelling
- ✓ XGBoost prediction
- ✓ Residual-based interaction scoring
- ✓ Classification threshold optimisation
- ✓ Visualisation and interpretation

Instead of relying solely on experimental observations, the model predicted the **expected additive response** from individual compounds. Comparing predicted and observed responses enabled the identification of genuine non-additive interactions. Although demonstrated using phenolic antioxidants, the framework is transferable to **drug discovery, formulation science, biotechnology, environmental chemistry, agriculture, food science, and other multicomponent chemical or biological systems**.

For me, this project illustrates how **experimental science, statistics and machine learning** can be combined to build general analytical tools, not just answer a single biological question. I am grateful to my co-authors for a fantastic collaboration and for bringing complementary expertise to this multidisciplinary project.

I enjoy working at the intersection of **experimental science, data science and artificial intelligence**, transforming complex datasets into practical, reproducible analytical workflows that generate actionable insights and support evidence-based decision making.

If you're interested in collaborating or exploring how machine learning could accelerate your research or product development, don't hesitate to contact me (delphine.vincent.consulting@gmail.com).



Publication

<https://www.nature.com/articles/s41598-026-63770-y>



Python workflow

https://github.com/dlf2024/Python_ML_antioxidant-assays_standards



Website portfolio:

<https://dlf2024.github.io/#phenolics-mixtures>



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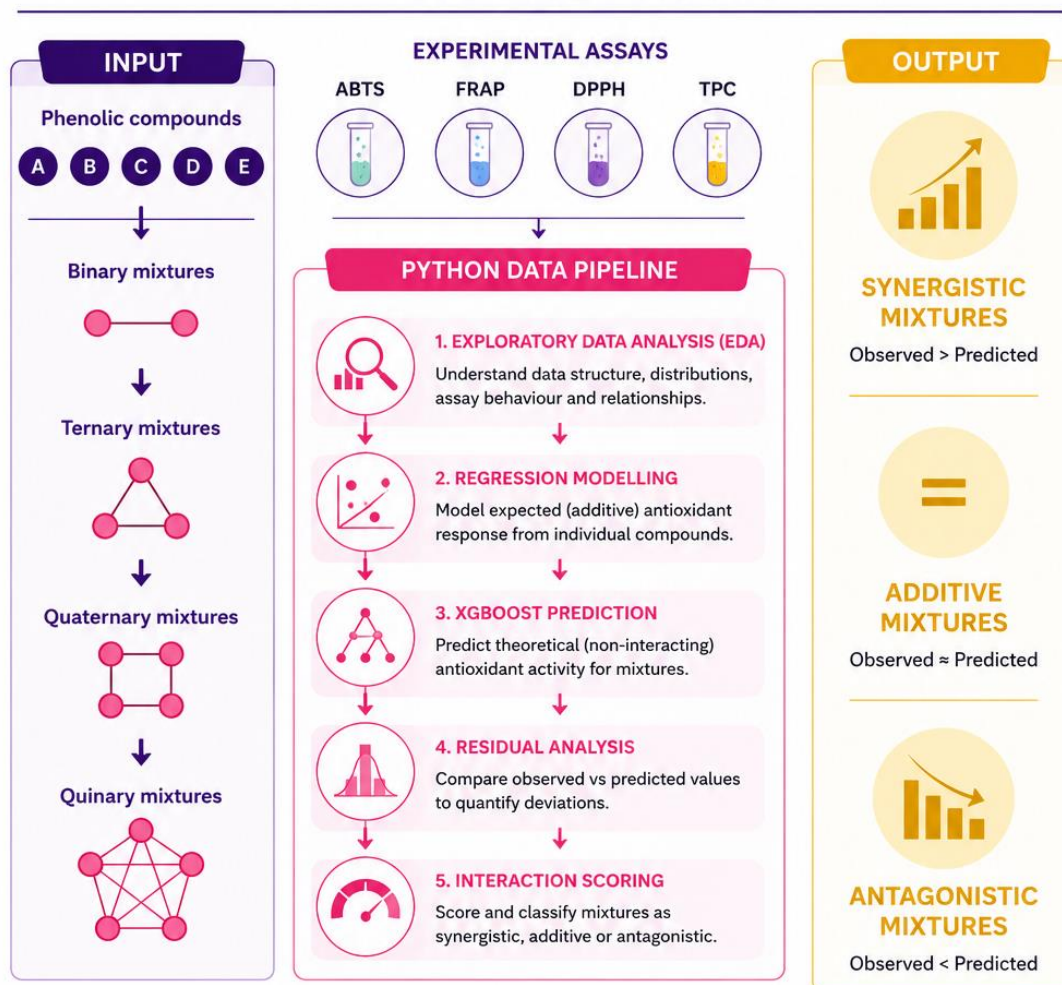


<https://dlf2024.github.io/>

SCIENTIFIC REPORTS 2026

Predicting Synergy and Antagonism of Phenolic Compound Mixtures

A machine learning framework to quantify non-additive interactions between phenolic compounds using antioxidant assays.



APPLICATIONS



Food science



Functional foods



Nutraceuticals



Biochemical research



Machine learning for analytical chemistry



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Read the paper:
<https://www.nature.com/articles/s41598-026-63770-y>



Python workflow:
https://github.com/dlf2024/Python_ML_antioxidant-assays_standards